Removing Ocular Artifacts using Wavelet Decomposition

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1 Comparing fastICA and Wavelet Decomposition

The goal is to compare the fastICA approach, which is the most common approach in the literature, with a wavelet transform-based approach for removing ocular artifacts from TMS-EEG data. The objective is to identify a method that preserves TEP information while eliminating ocular artifacts.

1.1 Dataset

`alberto8_tdcs_epoched.set`, which is a raw TMS-EEG dataset, epoched (`-1000` to `1000` ms) around the TMS pulse.

1.2 The pipeline

The pipeline I used for removing ocular artifacts using wavelet transform is the following:

1. Remove unused channels.
2. Add EOG events to the dataset (optional).
3. Remove bad channels.
4. Apply ocular artifacts removal method:
   (a) fastICA
   (b) Wavelet decomposition

The wavelet decoomposition method is applied on the whole signal, for this reason is not necessary to have the EOG events.

1.2.1 FastICA

To apply the fastICA method, the TESA matlab toolbox has been used. The following code snippet shows the used parameters and how to apply fastICA for ocular artifact removal:

```
% Highpass filter before ICA (0.1 Hz)
EEG = pop_tesa_filtbutter( EEG, 0.1, [], 4, 'highpass' );

% fastICA for ocular artifacts
EEG = pop_tesa_fastica( EEG, 'approach', 'symm', 'g', 'tanh', '
    stabilization', 'off' );
EEG = pop_tesa_compselect(EEG, ...
    'blink','on', ...
    'blinkThresh',2.5, ...
    'blinkFeedback','off', ...
    'move','on', ...
    'moveThresh',2, ...);
```

1.2.2 Wavelet Decomposition

To remove the ocular artifact has been used a daubechies wavelet 4 (db4) Fig. 1 with six levels of decomposition. In order to remove the artifact, the approximation coefficients (which is where most of the artifact reside) have been replaced with the mean value of the level.

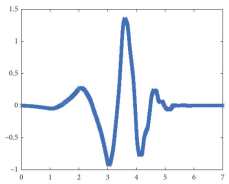


Figure 1: Real part of the Daubechies wavelet 4 (db4).

The following code snippet shows how I have applied the method in Matlab:

```
%% Wavelet Ocular Artifacts Removal
function EEG = wavelet_OA_rm(EEG)
    levels = 6; % number of levels
    wname = 'db4'; % wavelet name
    for chanIdx = 1:EEG.nbchan
        [c, l] = wavedec(EEG.data(chanIdx, :), levels, wname);
        base_value = mean(c(1:l(1)-1));
        c(1:l(1)-1) = base_value;
        EEG.data(chanIdx, :) = waverec(c, l, wname);
    end
```

1.3 Results

The results of the comparison between fastICA and wavelet transform in removing ocular artifacts.

1.3.1 Power Spectral Density (PSD) Comparison

The power spectral density has been calculated on the FP2 channel.

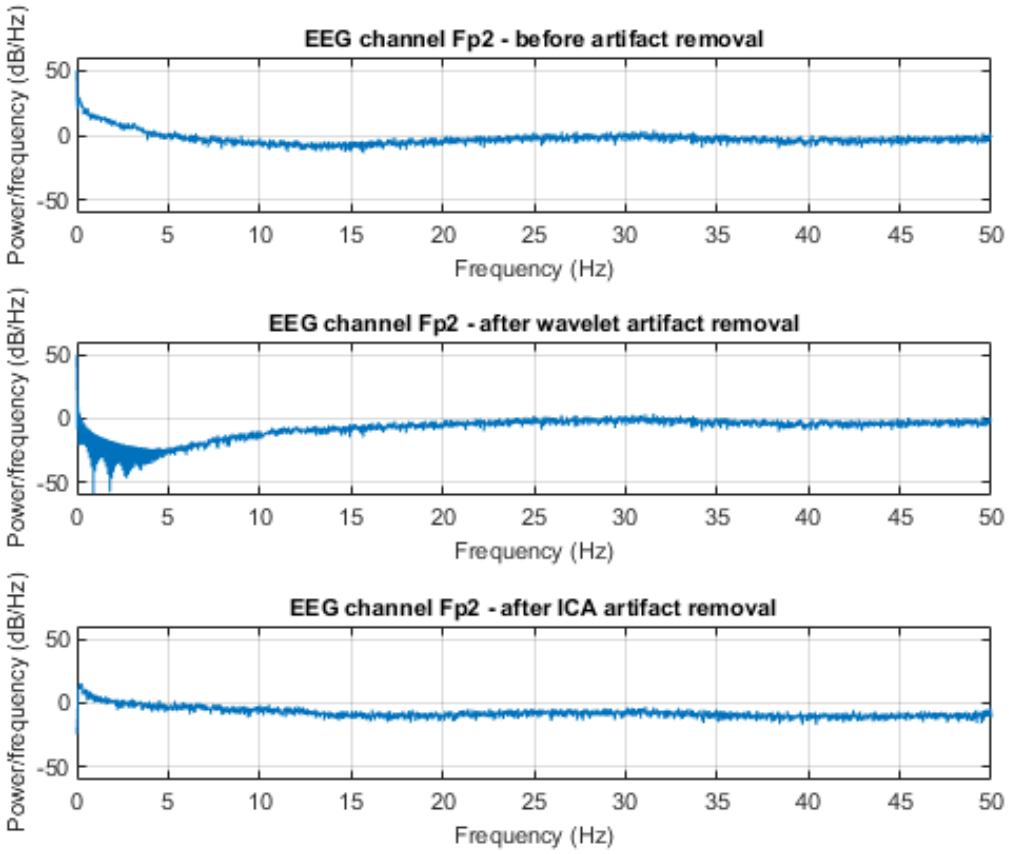


Figure 2: PSD graphical comparison before and after artifact removal.

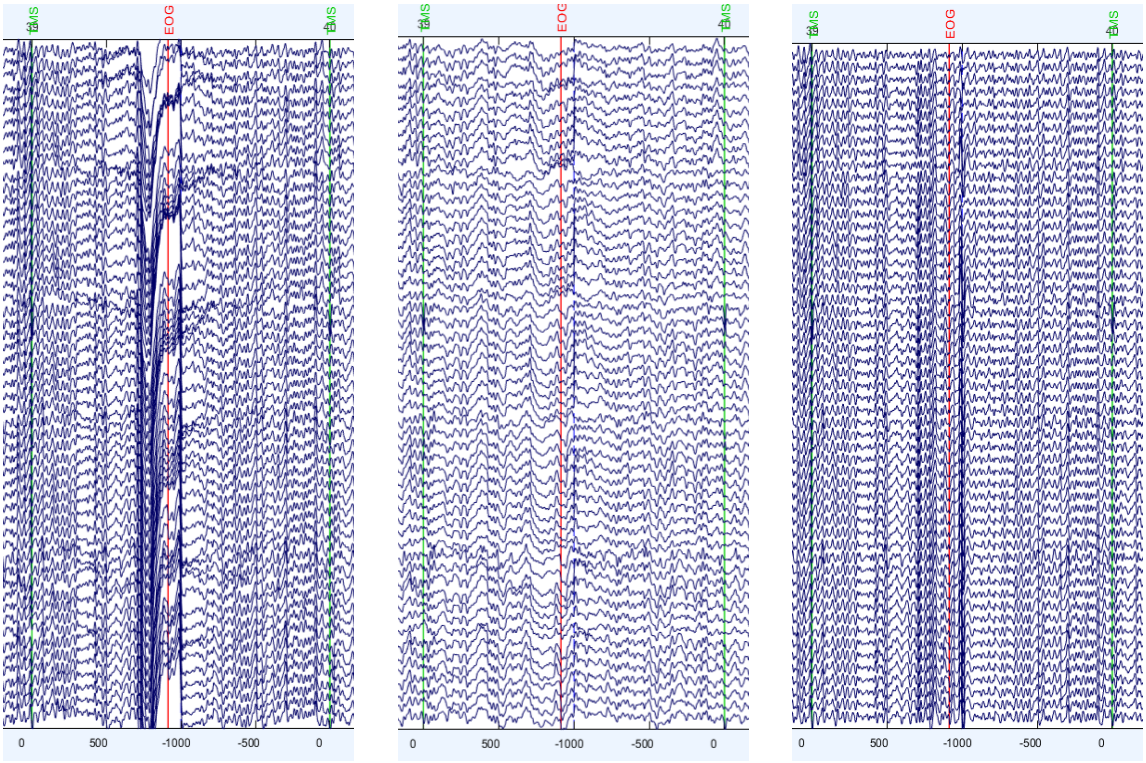
In order to compare the differences in the power spectral density the **Root Mean Square Error** has been calculated, with the following results:

- Wavelet vs fastICA: **40.98** dB/Hz
- Wavelet vs original: **12.93** dB/Hz
- fastICA vs original: **42.45** dB/Hz

The results show that the signal after wavelet transform filtering is closer to the orignal, in terms of frequency content, than the signal after fastICA filtering. And the two artifact removal methods have quite different results.

1.3.2 Visual Comparison

In order to visualize in a clearer way the data a bandpass (1-48 Hz) filter was applied. Figure 3a shows raw data with a clear ocular artifacts in epoch 38. Figure 3b shows the channel after fastICA ocular artifact removal, while figure 3c shows the channels after wavelet transform ocular artifact removal. Both methods show some residual artifacts.



(a) Raw data. (b) FastICA. (c) Wavelet decomposition.

Figure 3: Comparison of EEG channels (epoch 39-40): (a) raw, (b) after fastICA, (c) after wavelet decomposition.

1.4 Possible improvements

1. Use more decomposition layers or a different wavelet type, in order too isolate the artifact in a detail level instead of the approximation coefficient as it can lead to lose some TEP information.

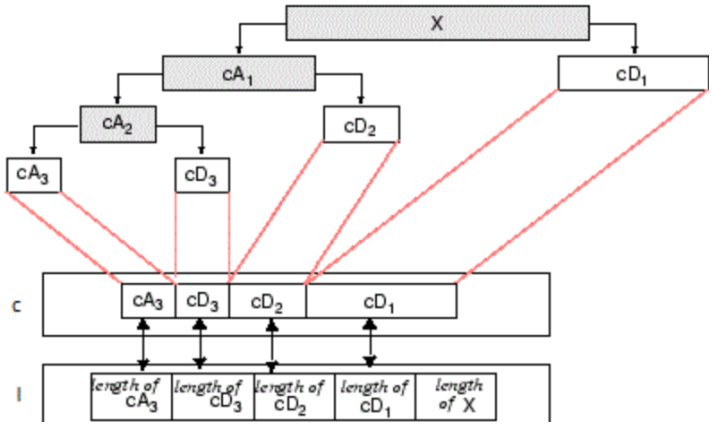


Figure 4: The image shows that the wavedec matlab function operates by dividing the signal into approximation coefficients (cA) and detail coefficients (cD). Then for eache level the approximation coefficients are divided again into approximation coefficients and detail coefficients.

2. Use a different method of removing the artifact from the approximation coefficients.

2 ICA vs Wavelet Decomposition vs EMD

2.1 Artifact removal methods

The following table summarizes the parameters used for the artifact removal methods applied to the TMS-EEG data.

Wavelet transform	Wavelet type	Daubechies 4 (db4)
	Levels of decomposition	12
	Detail levels removed	7,8,9,10,11,12
Empirical Mode Decomposition	Interpolation	pchip
	Sift Relative Tolerance	0.2
	Max Num IMF	5
	Sift Max Iterations	100
	Max Number Extrema	1
	Max Energy Ratio	20
	IMFs removed	3,4
Independent Component Analysis	Approach	symm
	g	tanh
	Blink Threshold	2.5
	Move Threshold	2

2.2 Results

The following figures show the results of the comparison between the three artifact removal methods.

2.2.1 Power Spectral Density (PSD) Comparison

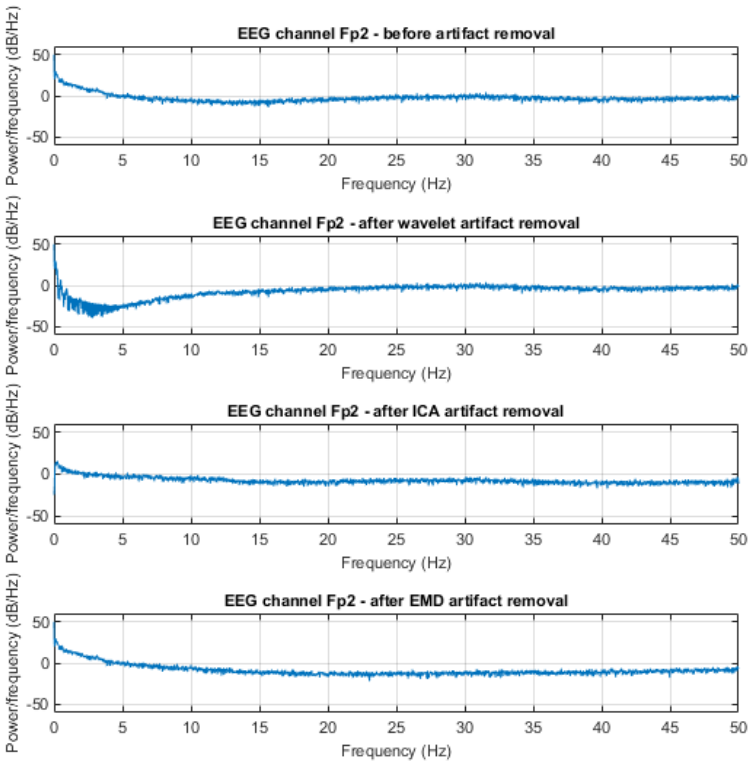


Figure 5: PSD graphical comparison before and after artifact removal.

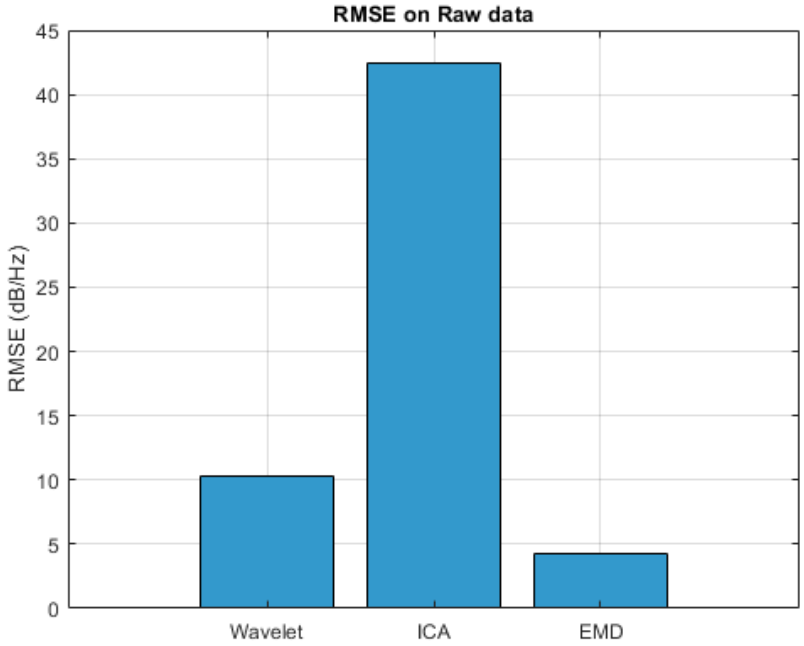
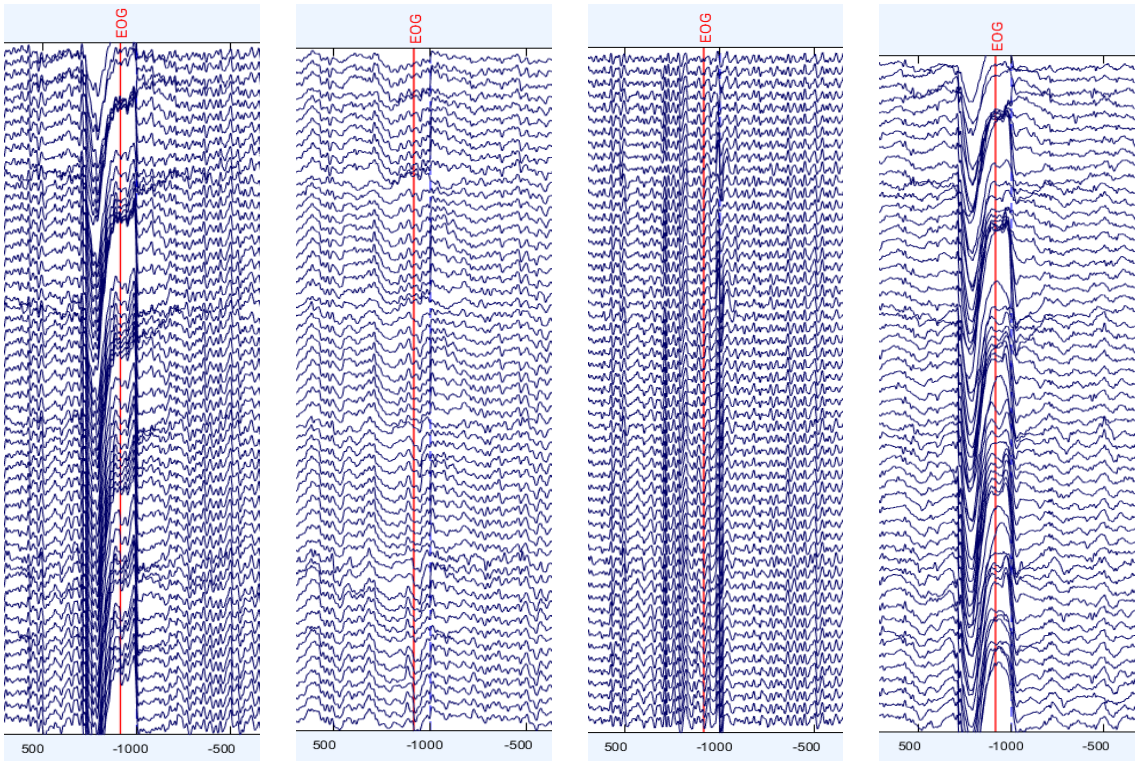


Figure 6: Bar plot of the Root Mean Square Error (RMSE) of the Power Spectral Density (PSD) for each method.

2.2.2 Visual Comparison



(a) Raw data. (b) FastICA. (c) Wavelet. (d) EMD.

Figure 7: Comparison of EEG channels (epoch 39-40): (a) raw, (b) after fastICA, (c) after wavelet decomposition, (d) after Empirical Mode Decomposition.

2.2.3 Removed signal comparison

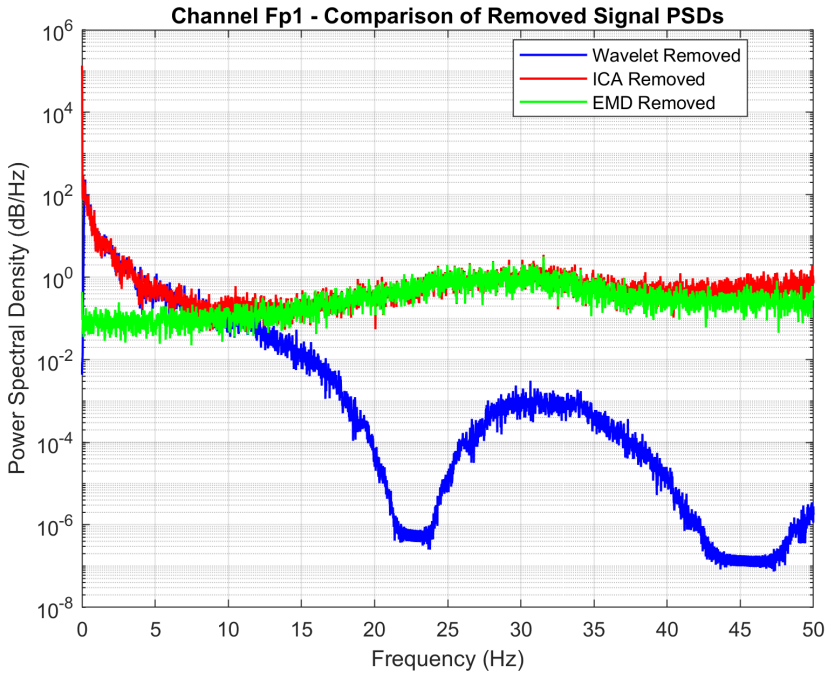
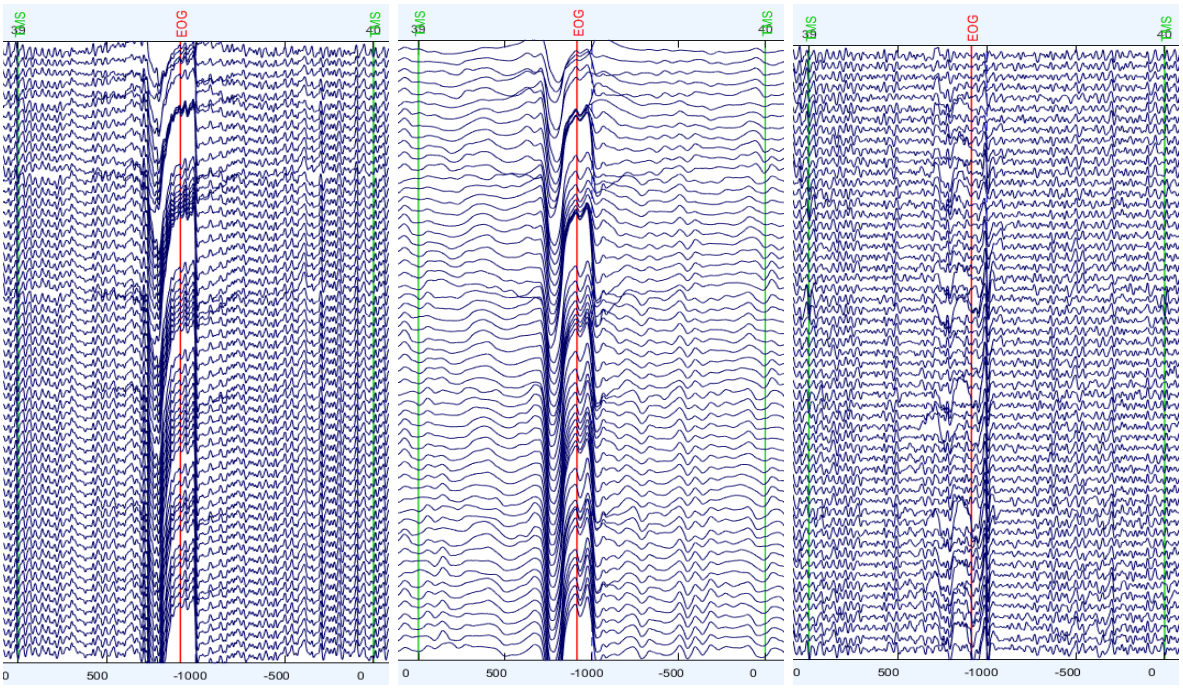
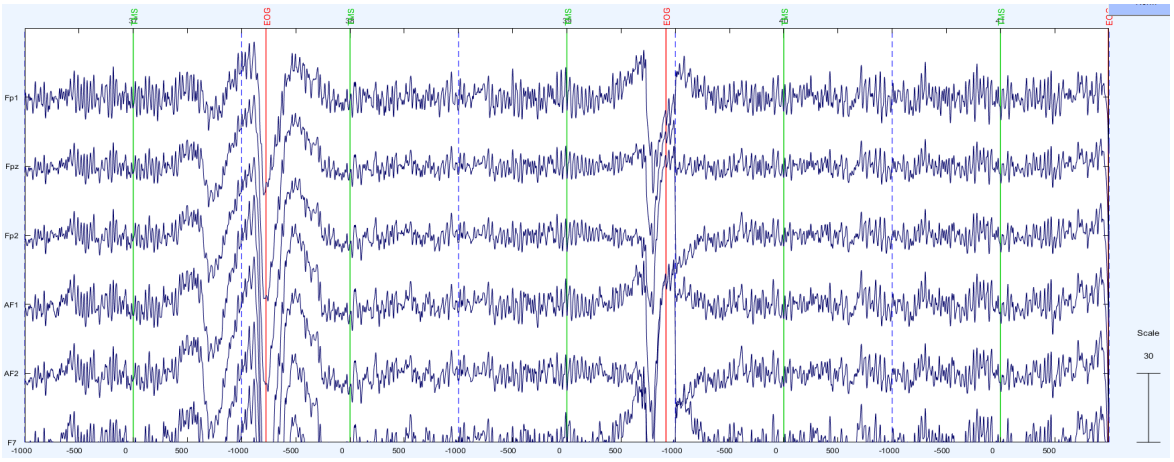


Figure 8: PSD of the removed components for each method.

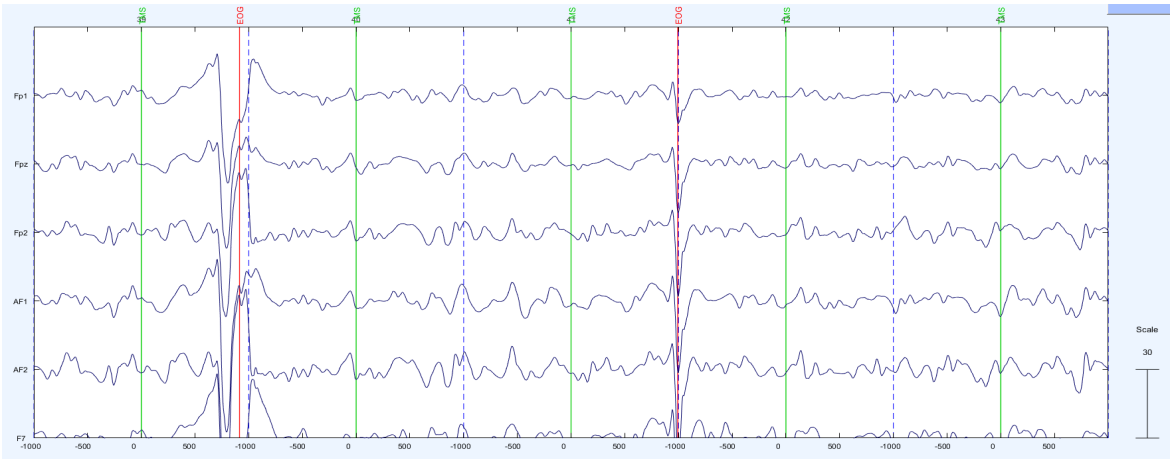


(a) fastICA. (b) Wavelet. (c) EMD.

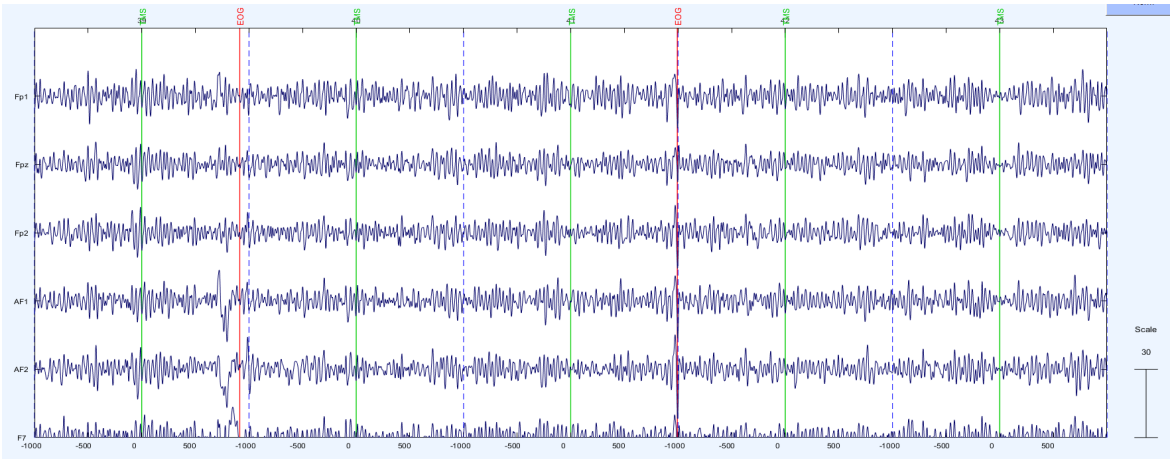
Figure 9: Comparison of the signal removed by the three methods (epoch 39-40): (a) after fastICA, (b) after wavelet decomposition, (c) after EMD.



(a) fastICA.



(b) Wavelet.



(c) EMD.

Figure 10: 5 channels comparison of the signal removed by the three methods: (a) after fastICA, (b) after wavelet decomposition, (c) after EMD.

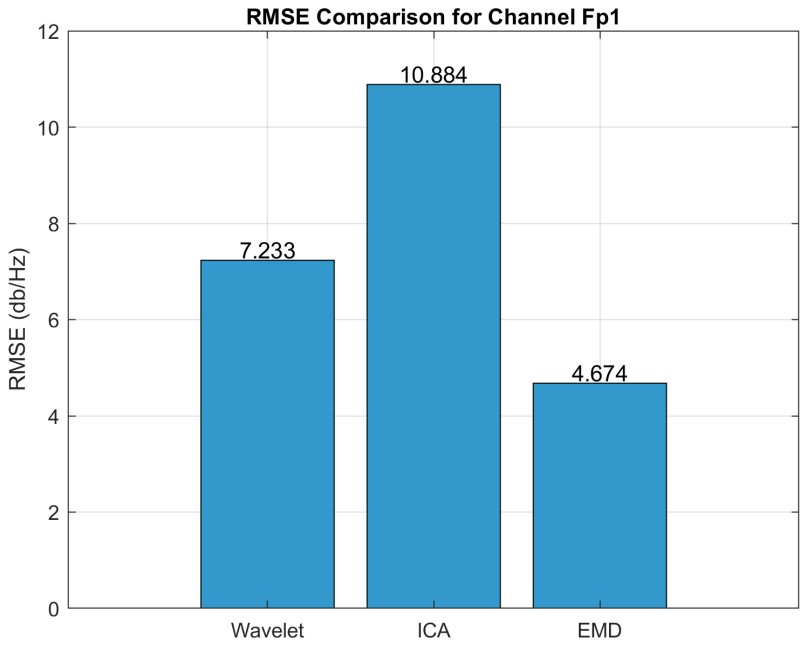


Figure 11: RMSE has changed, as a result of applying 0.1 Hz hipass filter before every method (before was applied only before fastICA, and was icreasing its rmse).